5CC512 Communication and Security Protocols

Security Basics

Instructor Contact & Resources

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How to succeed in the module

- **Attend all lectures and tutorials**
- Engage in the module and participate in discussions
- Seek early feedback on coursework
- Consider assessment criteria
- Be curious and motivated to learn
- Investigate academic literature
- **Communicate**

Why information security?

- Information has always had value through history
- Cryptography has been used for centuries to conceal valuable information
  - Hieroglyphs (4,000BC)
  - Atbash substitution cipher
  - Caesar shift cipher
  - Polybius square (170BC)
  - Mlecchita vikalpa
- **Why do we need cryptography?**
- **Why do we need information security?**

Difference between theory and practice

- "For the most part, cryptography has done little more than give Internet users a false sense of security by promising security but not delivering it. And that's not good for anyone except the attackers.
  The reasons for this have less to do with cryptography as a mathematical science, and much more to do with cryptography as an engineering discipline."

  Bruce Schneier, Practical Cryptography

- Making cryptography work in a practical system as part of an information security architecture is really important

- **Is cryptography alone enough?**

COMPUTER SECURITY CHALLENGES

- Security is not simple
- Potential attacks on the security features need to be considered
- Procedures used to provide particular services are often counter-intuitive
- It is necessary to decide where to use the various security mechanisms
- Requires constant monitoring
- Is too often an afterthought
- Security mechanisms typically involve more than a particular algorithm or protocol
- Security is essentially a battle of wits between a perpetrator and the designer
- Little benefit from security investment is perceived until a security failure occurs
- Strong security is often viewed as an impediment to efficient and user-friendly operation

OUR CONCERN

- Security Protocols
  - Some of them can be applied to Computer security
- What is protocol?

protocol *noun*

C1 OPAL W

/ˈprəʊtəkɒl/

/ˈprəʊtəkɑːl/

1 ★ C1 [uncountable] a system of fixed rules and formal behaviour used at official meetings, usually between governments
- *a breach of protocol*
- *the protocol of diplomatic visits*

2 ★ [countable] *(specialist)* the first or original version of an agreement, especially a treaty between countries, etc.; an extra part added to an agreement or treaty
- *the first Geneva Protocol*
- *It is set out in a legally binding protocol which forms part of the treaty.*
- *The dates were agreed under a protocol to the climate convention.*

TOPICS **Discussion and agreement** C1

3 ★ [countable] *(computing)* a set of rules that control the way data is sent between computers

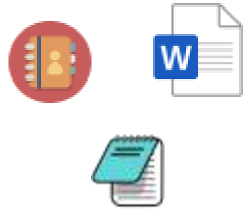
4 ★ [countable] *(specialist)* a plan for performing a scientific experiment or medical treatment

TOPICS **Scientific research** C2

www.oxfordlearnersdictionaries.com

Computer Security Concepts

Physical Security



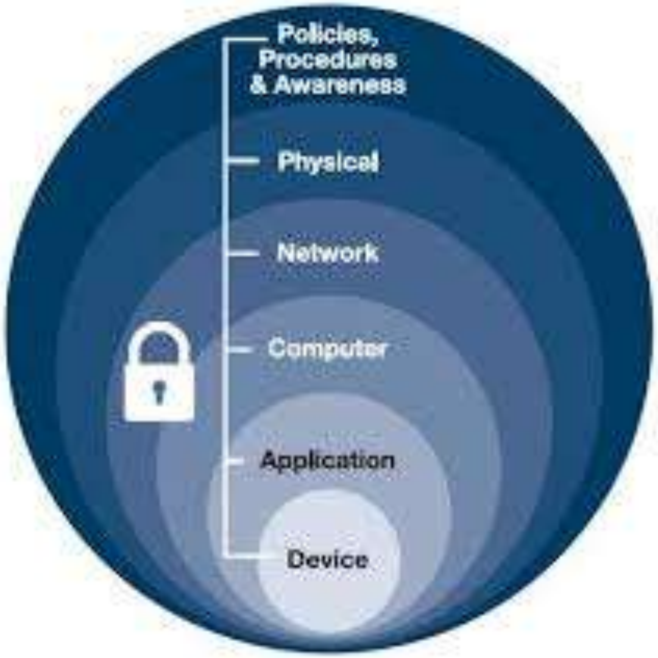
Computer Security

Network Security

Computer Security Concepts

- What is our strategy considering this fact?

- Defense in Depth



Computer Security Objectives

Confidentiality

- Data confidentiality
  - Assures that private or confidential information is not made available or disclosed to unauthorized individuals
- Privacy
  - Assures that individuals control or influence what information related to them may be collected and stored and by whom and to whom that information may be disclosed

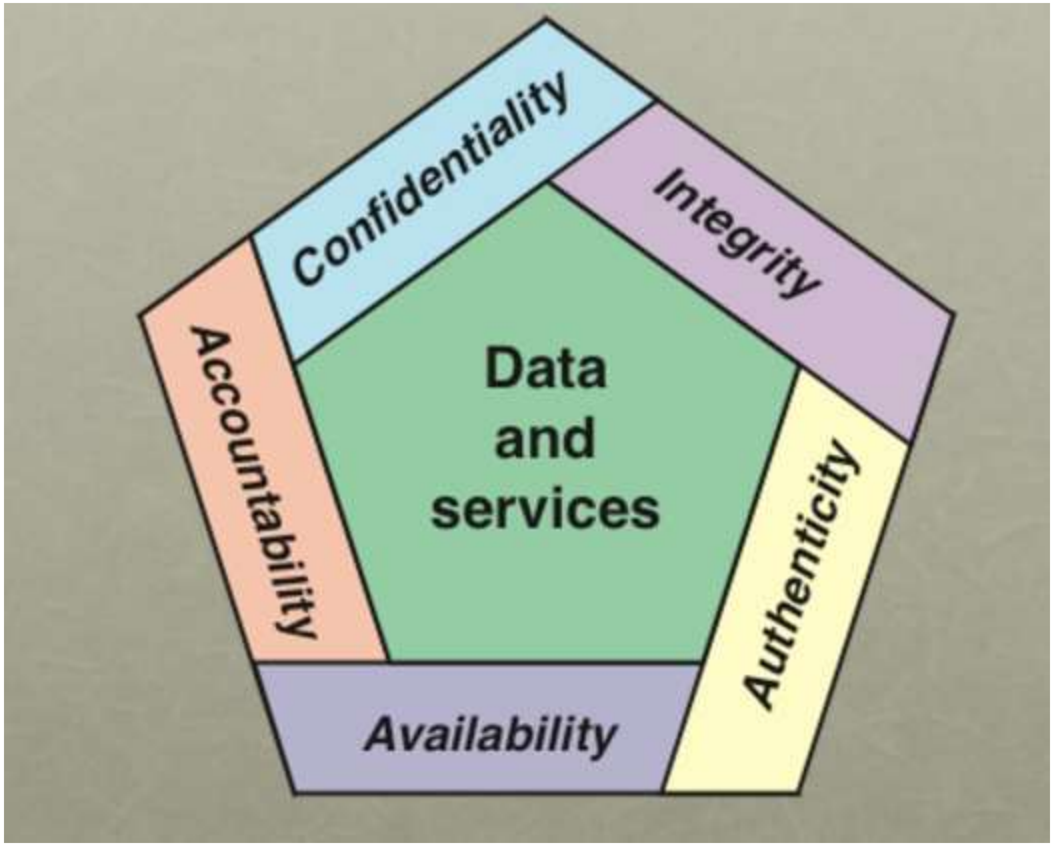
Integrity

- Data integrity
  - Assures that information and programs are changed only in a specified and authorized manner
- System integrity
  - Assures that a system performs its intended function in an unimpaired manner, free from deliberate or inadvertent unauthorized manipulation of the system

Availability

- Assures that systems work promptly and service is not denied to authorized users

CIA TRIAD


Confidentiality
Integrity
Data
and
services
Accountability
Authenticity
Availability

POSSIBLE ADDITIONAL CONCEPTS

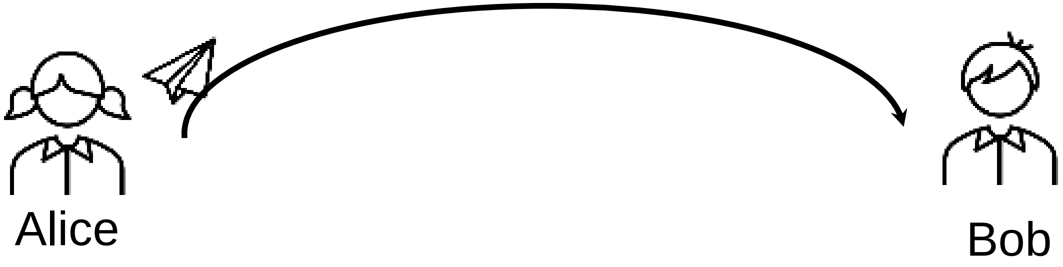
Authenticity

- Verifying that users are who they say they are and that each input arriving at the system came from a trusted source

Accountability

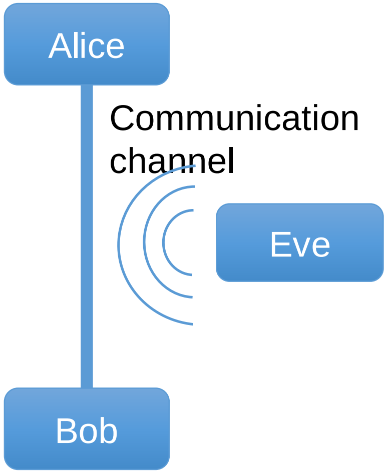
- The security goal that generates the requirement for actions of an entity to be traced uniquely to that entity

Example scenario: Message transfer



- **Alice would like to send a message to Bob**
- **What questions might they ask about the security of the message exchange?**

Cryptography – Security Services



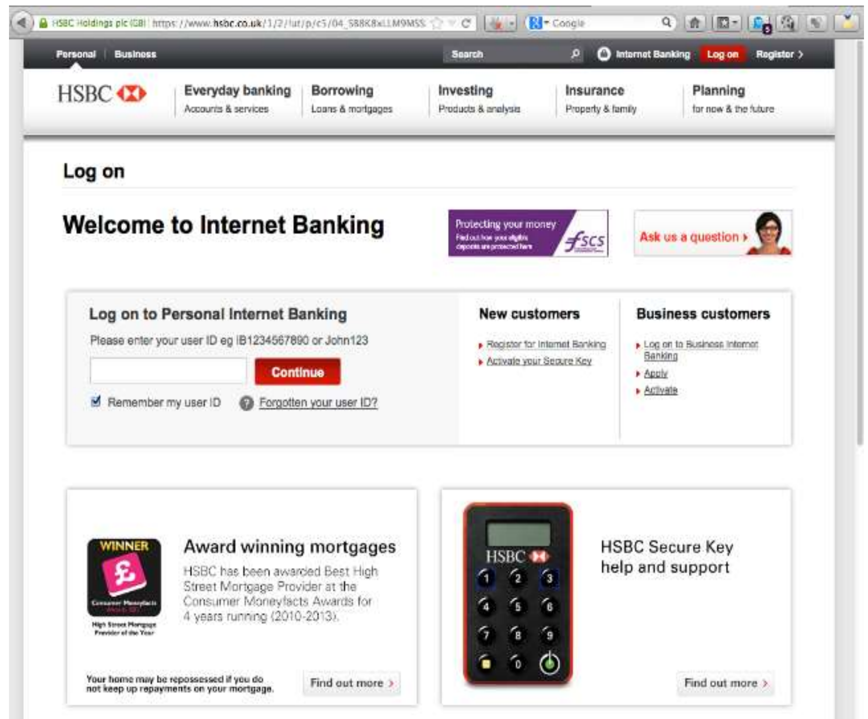
- **Security services -** Security goals we would like to achieve
  - **Confidentiality** – assurance that data cannot be viewed by unauthorised entities
  - **Data Integrity** – assurance that data <u>has not been</u> altered in unauthorised manner
  - **Data origin authentication** – assurance a given entity was the original source of received data
  - **Non-repudiation** – Assurance an entity cannot deny a previous action
  - **Entity authentication –** assurance an entity is involved and <u>currently active</u> in a communication
- Consider relationships between these services

Cryptography is everywhere

- **To secure our communications**
  - Web traffic (HTTPS), Wireless traffic, (WEP, WPA), …etc.
- **User authentication**
- **Content protection** (e.g. DVD): CSS,AACS
- **Encryption of files**: EFS, TrueCrypt
- **Electronic Voting**
- **Credit cards**
- …

Where do we use cryptography ?

Internet: Is this really the official website of my bank ?



Where do we use cryptography ?

Electronic Voting:

- Is the vote confidential?
- How authorised people can be authenticated to the system and vote?
- Is the vote 100 % correct?

Weekly plan

Lecture	Week	
1	9	**Security Basics**
2	10	**Symmetric Encryption and Message Confidentiality**
3	11	**Modern Algorithms for Symmetric Encryption (AES, DES)**
4	12	**Message Authentication - Hash Function**
5	13	**Public Key Cryptography**
6	14	**User Authentication**
7	15	**Key Distribution: Kerberos & Public Key Distribution**
8	16	**Email Security: SMTP, IMAP**
9	17	**Enhanced Email Security (PGP, S/MIME)**
10	18	**Transport Layer Security: TLS, SSH**
11	19	**HTTPS & Secure Shell (SSH)**
12	20	

Any Questions?